

Comparison of 1D and 2D CNN for the predictive maintenance of Bearing Faults using Deep Learning Methods*

Rajshri Simhadri^[0009–0000–2789–9958], Vidushi Agarwal^[0000–0001–6164–2240],
Kshirasagar Naik^[0000–0002–5083–2599], and Mahesh Pandey^[0000–0002–1064–4905]

University of Waterloo, Canada

Department of Electrical and Computer Engineering

{rajshri.simhadri, vidushi.agarwal, snaik}@uwaterloo.ca

Department of Civil and Environmental Engineering

mdpandey@uwaterloo.ca

Abstract. Rotating machinery, such as induction motors, forms the backbone of many industrial plants and processes. Bearing faults caused by high-speed motion and associated mechanical stress are the most dominant factors that disrupt the operation of motors and adversely affect the safety and performance of the plant. Therefore, a predictive maintenance to minimize the unplanned downtime of the system is required. This paper presents a comprehensive deep learning-based framework for bearing fault diagnosis using motor vibration data. The proposed approach involves the transformation of measured vibration signals into two-dimensional spectrograms using the short-time Fourier transform (STFT). Then, the paper explores three neural architectures: (1) a custom Dilated 1D Convolutional Neural Network (CNN), and (2) a pre-trained XceptionNet71 model (3) a pretrained EfficientNet V2-M model. The bearing fault dataset published by Case Western Reserve University (CWRU) is utilized for training and cross-validation of the two models. In practical deployments, low-cost IoT (Internet of Things) vibration sensors mounted on operational motors can continuously collect real-time data and transmit it over industrial networks to edge or cloud servers, where the proposed deep learning framework can analyze signals and flag emerging faults. Results of the analysis demonstrate that a 2D-channel STFT representation combined with a pretrained CNN significantly improves accuracy over the traditional signal-based models. This work demonstrates the effectiveness of fusing signal processing with deep convolutional models and provides a scalable solution for intelligent condition monitoring of motor bearings.

Keywords:

Induction Motors · Bearing Faults · Anomaly Detection · Short-Time Fourier Transform · Convolutional Neural Networks ·

* IFToMM Rotordynamics Template

1 Introduction

Fault detection for rotating machinery plays a crucial role in industrial applications. Rolling element bearings, in particular, are critical components that support loads, reduce friction, and ensure stable operation. Despite their mechanical simplicity, bearings are highly susceptible to wear, localized defects, and progressive degradation. The increasing adoption of industrial platforms [1] has further elevated the need for automated, data-driven, and real-time fault diagnosis methods [2]. Among available sensing modalities, vibration analysis remains one of the most effective techniques for capturing transient impulses and periodic modulations associated with early-stage bearing failures. Within this context, the CWRU bearing fault dataset has become a widely used benchmark for evaluating deep-learning architectures. It provides vibration recordings from bearings with single-point defects of varying severities on the inner race, outer race, and rolling element, along with a single healthy condition. However, the experimental design introduces a strong imbalance toward faulty conditions, which does not reflect real-world environments where healthy bearings are significantly more prevalent.

Although the CWRU dataset has been extensively studied, its ability to support fault classifiers that generalize to different operating environments is still not well established. While prior studies have highlighted several limitations in about CWRU dataset which are segment-level leakage from random splitting and bearing-identity leakage in load-based splits these issues have already been documented and addressed by researchers. Building on this understanding, the work establishes a unified and reproducible evaluation framework that applies a consistent segmentation, enabling magnified visibility of transient fault signatures. The framework enables a fair comparison of 1D, 2D, and 3D CNN architectures by adopting a consistent preprocessing pipeline that utilizes both standard STFT spectrograms and an extended differential representation (Delta-STFT) to enhance temporal sensitivity. To counteract the dataset’s class imbalance, targeted data augmentation has been applied to increase sample diversity, and transfer learning has been applied by fine-tuning an ImageNet-pretrained XceptionNet-71 and Efficient-Net V2-M, model while evaluating the safety-critical misclassification to interpret the results. These design choices strengthen the comparability of the CNN models, and practical interpretability of deep-learning-based bearing fault diagnosis within the dataset’s constraints. The previously identified challenges are addressed by our proposed approach as follows:

- We mitigate the issue of data imbalance by we apply weighted loss function during training, assigning higher penalties to underrepresented fault classes. This improves the model’s sensitivity to minority fault patterns and ensures balanced classification performance under varying operating conditions.
- To enhance the detection of subtle and noise-obscured fault signatures, we leverage both spatial and temporal information present in vibration signals through a hybrid feature extraction approach. Specifically, 1D CNN’s capture temporal dependencies from raw time-domain sequences, while 2D CNN’s

operate on time-frequency spectrograms to extract spatial patterns, the 3D CNNs process stacked spectrograms or multi-channel representations, allowing convolutional kernels to capture spatiotemporal dependencies across time, frequency, and depth.

2 Related Work

Convolutional Neural Networks (CNNs) remain the dominant approach for spectrogram based bearing fault diagnosis due to their ability to learn localized fault patterns from time–frequency representations. Prior studies have improved CNN performance under noise and varying loads through spectrogram preprocessing, denoising, and feature-preservation strategies. As cited in, STFT-based spectrograms which is effective even under low-SNR conditions [3], while wavelet-based pooling layers within residual CNNs help retain low-frequency fault components and enhance robustness [4]. Other works integrate CNNs with adaptive decomposition methods such as EMD, VMD, or blind deconvolution to suppress noise and strengthen discriminative features [5, 6].

Despite their strong performance, hybrid architectures that outperform CNN baselines introduce trade-offs, which includes increased computational cost and a greater dependence diverse datasets. Addressing challenges related to efficiency, interpretability, and deployment feasibility therefore remains essential for real-world adoption. A key challenge arises from the nature of benchmark datasets such as CWRU, which contain multiple recordings but only a small number of physical bearings per fault class. Even after segmentation, recordings originating from the same bearing inevitably appear across different subsets, making it difficult to separate bearing-specific characteristics from true fault signatures. This inherent overlap can lead to performance estimates that are optimistic relative to deployment scenarios. Additionally, many studies provide limited discussion of deployment-oriented indicators such as inference latency, computational cost, model size, or false-alarm tendencies, leaving open questions regarding suitability for real-time or embedded monitoring systems.

Beyond these concerns, the literature remains largely centered around single-resolution STFT representations, with few studies examining how variations in model dimensionality (1D, 2D, or 3D CNNs) can be evaluated fairly. Extended STFT variant, del-STFT has been investigated in this work. A unified fine-tuning strategy for ImageNet-pretrained models was adopted. This standardization enhances the data efficiency and convergence stability.

3 Methodology

Bearing failures are one of the leading causes of breakdowns in rotating machinery, often resulting in costly downtime, reduced efficiency, and even catastrophic system damage if not detected early. Early fault detection and classification are therefore critical to ensure the reliability and safety of industrial systems. Among the various diagnostic approaches, vibration-based monitoring is widely

regarded as the most effective, since fault signatures manifest as distinct patterns in vibration signals that evolve with fault type and severity.

3.1 Dataset and Signal Acquisition

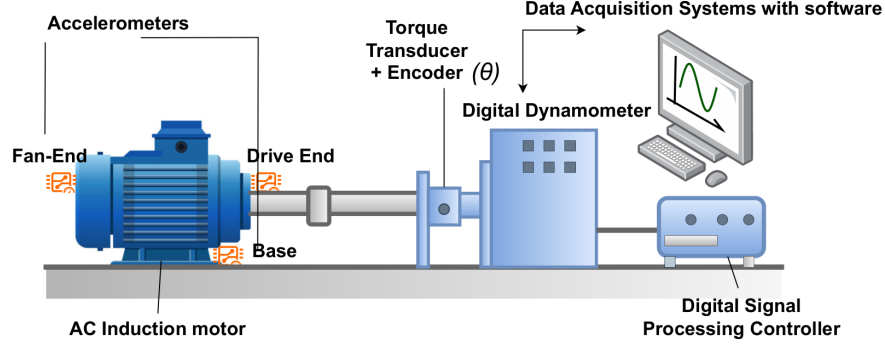


Fig. 1: CWRU bearing fault data acquisition setup

The CWRU Bearing fault dataset [?] is a collection of experiments that consists of experiments as shown in Figure 1 where a three-phase induction motor coupled to a torque transducer and a controlled dynamometer load. Vibration signals were measured using accelerometers mounted at the drive end (DE) and fan end (FE) under four motor loads (0 to 3 HP). The faults were introduced artificially using Electrical Discharge Machining (EDM) on three bearing components: the ball, inner race, and outer race. For each location, three defect diameters (0.007", 0.014", and 0.021") represent increasing severity levels.

In most cases, the experiments used a 12 kHz sampling rate, while some used 48 kHz. All the experiments consist of signals that are approximately 10 seconds long. Measurements with a sampling rate of 48 kHz were resampled to 12kHz. The table 1 summarizes the multi-fault classification configuration considered in this paper (specified by fault location, type and severity level/fault diameter measured in inches)

3.2 Problem Formulation

Bearing fault detection and diagnosis can be framed in several different ways, this impacts how a model predicts and the effectiveness is measured. Considering the vibration signals collected from the rotating machine which represent the dynamic response of the system under healthy and faulty operating conditions.

This helps us to extract discriminative fault signatures, each time-domain segment $\mathcal{S}$ is transformed into a time-frequency representation using the Short-Time Fourier Transform (STFT), yielding a spectrogram X_{ij} . This feature rep-

Table 1: Fault type and severity mapping used in this study (CWRU dataset).

Class ID	Fault Type	Diameter (in)
0	Ball Fault	0.007
1		0.014
2		0.021
3	Inner Race Fault	0.007
4		0.014
5		0.021
6	Outer Race Fault	0.007
7		0.014
8		0.021
10	Normal (Healthy)	—

resentation is then passed to a deep learning model M , which maps the input to a predicted class label:

$$\hat{y} = M(X_{ij}), \quad (1)$$

where $\hat{y} \in \{0, 1, \dots, K_{\text{cls}} - 1\}$ represents the model’s classification output.

The objective of this study is to develop a robust CNN-based framework capable of jointly detecting fault type and fault severity from non-stationary vibration signals. This enables early and reliable fault identification to support predictive maintenance.

3.3 Multi-label fault diagnosis approach

The time-series signals collected from the accelerometers $\mathcal{S}$ are initially at two different sampling frequencies (12 kHz and 48 kHz), all signals were uniformly resampled to $\mathcal{S}_r$ to a common rate of f_s^{target} 12 kHz to ensure consistency across operating conditions. Now, each resampled signal is divided into fixed-length 5 s windows (60,000 samples) with a 1 s hop (12,000 samples). This segmentation ($\{x_{ij}\}$) increases sample density and preserves temporal continuity, ensuring that transient fault-related impulses are retained within at least one analysis window.

Given the limited size and class imbalance of the CWRU dataset, five segment-level augmentation strategies were applied to expand the training distribution. Gaussian noise with a very small standard deviation was added to emulate sensor and ambient disturbances, and amplitude scaling was applied by multiplying each segment by a slightly varied random factor close to one to mimic changes in sensor sensitivity and operating load. Time translation was introduced by circularly shifting each segment to provide phase variability and reduce dependence on strict temporal alignment, while random masking was performed by zeroing out roughly ten percent of the samples to imitate signal dropouts or temporary wiring issues. Finally, small temporal perturbations were incorporated to create slight, realistic inconsistencies in sampling while preserving the underlying periodic nature of the fault signatures.

After segmentation and augmentation, each signal segment is transformed into the time–frequency domain using the Short-Time Fourier Transform (STFT)

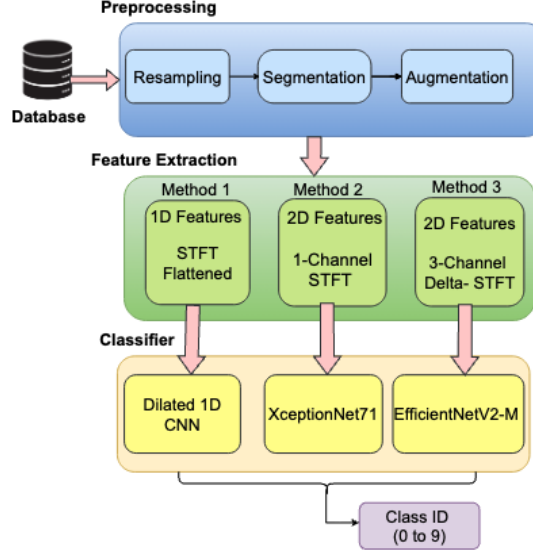


Fig. 2: Workflow of experiments

with a Hann window and 75% overlap. The resulting complex coefficients are converted into magnitude-based power spectrograms X_{ij} as defined in (2), which are then log-scaled as per (3) to enhance contrast and compress dynamic range. The spectrograms are subsequently mean-variance normalized to standardize the input distribution, and finally resized to a fixed resolution suitable for CNN-based models.

$$X_{ij}(\ell, \omega) = |\text{STFT}(x_{ij})|^2, \quad (2)$$

$$X_{ij}(\ell, \omega) \leftarrow \log(1 + X_{ij}(\ell, \omega)), \quad (3)$$

3.4 Model Selection and evaluation methodology

The objective of this study is to develop a robust CNN-based framework capable of jointly detecting fault type and fault severity from non-stationary vibration signals. This enables early and reliable fault identification to support predictive maintenance. The work focuses on an efficient feature-extraction pipeline based on STFT representations and the strategic use of transfer learning to improve model performance.

3.5 Model architectures and training details

CNN-based models are widely used in bearing fault diagnosis due to their strong feature extraction capability. This study evaluates three representative architectures under a unified preprocessing and CVM-CV training protocol: a Dilated

Algorithm 1: Inference Pipeline for Bearing Fault Classification

Input: Raw vibration signals $\mathcal{S}$, target sampling rate f_s^{target} , pretrained XceptionNet-71 model M

Output: Predicted bearing fault label $\hat{y}$

Function Resample($\mathcal{S}$)

- foreach** $s_i \in \mathcal{S}$ **do**
- Resample s_i to f_s^{target} to obtain $s_{i,r}$;
- return** $\mathcal{S}_r = \{s_{i,r}\}$;

Function Segment($\mathcal{S}_r$)

- foreach** $s_{i,r}$ **do**
- Divide into overlapping windows x_{ij} (5 s window, 1 s hop);
- return** $\{x_{ij}\}$;

Function STFT_Features($\{x_{ij}\}$)

- foreach** x_{ij} **do**
- Compute STFT magnitude (2);
- Apply log compression and normalization (3);
- Resize to $X_{ij} \in \mathbb{R}^{128 \times 64 \times 1}$;

Function Infer($M, \{X_{ij}\}$)

- foreach** X_{ij} **do**
- $\hat{y}_{ij} \leftarrow M(X_{ij})$;
- return** $\text{MajorityVote}(\{\hat{y}_{ij}\})$;

Main Routine:

- $\mathcal{S}_r \leftarrow \text{Resample}(\mathcal{S})$;
- $\{x_{ij}\} \leftarrow \text{Segment}(\mathcal{S}_r)$;
- $\{X_{ij}\} \leftarrow \text{STFT_Features}(\{x_{ij}\})$;
- $\hat{y} \leftarrow \text{Infer}(M, \{X_{ij}\})$;
- return** $\hat{y}$;

1D-CNN for raw signals, an ImageNet-pretrained Xception71 for STFT spectrograms, and an ImageNet-pretrained EfficientNetV2-M for Delta-STFT spectrograms. Training configurations including learning rate, batch size, gradient accumulation, LR scheduling, and early stopping were optimized for stable convergence across folds. All models used the Adam optimizer and CrossEntropy-Loss, with pretrained 2D CNNs fine-tuned using domain-specific spectrogram inputs. A complete summary of the architecture-specific parameters is provided in Table 2.

4 Experiments and Results

This work presents a study on vibration-based bearing fault type and severity detection for predictive maintenance applications. Among the evaluated models, the ImageNet-pretrained XceptionNet-71, when fine-tuned, achieved the best

Table 2: Hyperparameter Settings for Each Experiment

Parameter	Exp 1: Dilated 1D CNN	Exp 2: Xception71	Exp 3: EfficientNetV2-M
Input Type	Raw 1D Signal	STFT Spectrogram	Delta-STFT Spectrogram
Input Shape	8192 samples	224 × 224	224 × 224
Input Channels	1	1 (Grayscale)	3 (RGB)
Architecture	Custom Dilated 1D CNN	xception71 (pretrained)	tf_efficientnetv2_m (pretrained)
Loss Function	CrossEntropyLoss	CrossEntropyLoss	CrossEntropyLoss
Optimizer	Adam	Adam	Adam
Learning Rate	1e-4	1e-4	1e-4
Weight Decay	1e-5	1e-5	1e-5
Batch Size	16 (AMP Enabled)	64 (Grad Accum)	64 (Grad Accum)
Gradient Accumulation	No	Yes (8 steps)	Yes (8 steps)
LR Scheduler	ReduceLROnPlateau (factor 0.5)	Same	Same
Epochs per Fold	10	10	10
Early Stopping	Yes (patience = 10)	Yes	Yes
Cross-Validation	10-fold Stratified K-Fold	Same	Same
Evaluation Metric	Macro F1-Score	Macro F1-Score	Macro F1-Score
Device	NVIDIA RTX 3060 Ti	Same	Same

overall performance, with a macro F1-score of 0.9979 and a mean ROC-AUC of 0.99 across all classes.

Table 3: Cross-Validation and Final Test Performance Summary

Model	Val Accuracy (Mean ± Std)	Val Macro F1 (Mean ± Std)	Test Macro F1
Dilated 1D CNN	98.61% ± 0.61%	0.9870 ± 0.0083	0.9908
XceptionNet71	98.97% ± 0.65%	0.9852 ± 0.0082	0.9979
EfficientNetV2-M	97.97% ± 0.71%	0.9796 ± 0.0088	0.9942

The proposed approach addresses ten fault classes using both signal processing techniques and spectrogram-based image features. Notably, the 2D channel spectrogram input demonstrated better class separability than flattened STFT and three-channel spectrogram representations.

Among the three models from the table as reported [3], XceptionNet71 achieved the highest final test set macro F1-score of 0.9979, closely followed by EfficientNetV2-M at 0.9942 and Dilated 1D CNN with 0.9908. The pretrained XceptionNet71 model demonstrated the highest overall performance with a final test macro F1-score of **0.9979** and excellent class-wise balance, as reflected in its classification report. It consistently achieved high precision and recall across all classes, with minimal performance variance across folds.

5 Conclusion

This work presented a study on vibration-based bearing fault classification using the Case Western Reserve University (CWRU) dataset. Three deep learning

architectures: a custom Dilated 1D-CNN, XceptionNet-71, and EfficientNetV2-M, were evaluated using STFT-based input representations to examine their ability to learn fault-related features. Class imbalance was addressed by employing a weighted loss function during training, which helped improve recognition of underrepresented fault classes. The framework also explored complementary temporal, spatial, and spatiotemporal feature learning through 1D, 2D, and 3D CNN approaches. The results indicate that combining imbalance-aware training with multi-dimensional CNN feature extraction can support reliable bearing fault classification under practical diagnostic conditions.

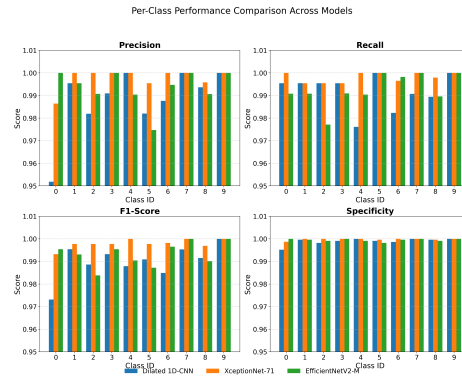


Fig. 3: Per-class performance comparison of Dilated 1D-CNN, XceptionNet-71, and EfficientNetV2-M across Precision, Recall, F1-Score, and Specificity

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